FIITJEE

ALL INDIA TEST SERIES

FULL TEST – V

JEE (Main)-2020

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A

Sol. When shell strikes the ball and gets stuck with it, combined body of mass 2m starts to move to the right. Let velocity of combined body (just after collision) be v_1.

According to law of conservation of momentum,

$(m + m) v_1 = mv_0$ or $v_1 = \frac{v_0}{2} = 3\ ms^{-1}$.

As soon as the combined body starts to move rightwards, thread becomes inclined to the vertical. Horizontal component of its tension retards the combined body while trolley accelerates rightwards due to the same component of tension.

Inclination of thread with the vertical continues to increase till velocities of both (combined body and trolley) become identical or combined body comes to rest relative to trolley. Let velocity at that instant of maximum inclination of thread be v. According to law of conservation of momentum.

$(2m + M) v = 2m.v_1$ or $v = 1\ ms^{-1}$

During collision of ball and shell, a part of energy is lost. But after that, there is no loss of energy. Hence, after collision, kinetic energy lost is used up in increasing gravitational potential energy of the combined body.

If maximum inclination of thread with the vertical be □ then according to law of conservation of energy,

$\frac{1}{2}(2m) v_1^2 - \frac{1}{2}(2m + M) v^2$

$= 2mg (\ell - \ell \cos\theta)$

$\therefore \quad \cos\theta = 0.8$ or $\theta = 37°$

2. A

Sol. The torque is defined as

$\tau = r \times F$

$$r \times F = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 7 & 3 & 1 \\ -3 & 1 & 5 \end{vmatrix}$$

$$= \hat{i}\begin{vmatrix} 3 & 1 \\ 1 & 5 \end{vmatrix} + \hat{j}\begin{vmatrix} 1 & 7 \\ 5 & -3 \end{vmatrix} + \hat{k}\begin{vmatrix} 7 & 3 \\ -3 & 1 \end{vmatrix}$$

$= \hat{i}(15 - 1) + \hat{j}(-3 - 35) + (7 - (-9))$

$= 14\hat{i} - 38\hat{j} + 16\hat{k}$

Thus the answer is (1)

3. D

Sol. The fringe–width β is given by

$$\beta = \frac{\lambda D}{2d} \quad \text{and} \quad \beta' = \frac{\lambda D'}{2d}$$

where λ is the wavelength of light used, D is the distance of the screen from the two slits and 2 d is the separation between two slits.

Now $$\beta - \beta' = \frac{\lambda(D - D')}{2d}$$

$$\Rightarrow \quad \lambda = \frac{(\beta - \beta')2d}{D - D'}$$

$$\lambda = \frac{(3\times10^{-5})(10^{-3})}{5\times10^{-2}}$$

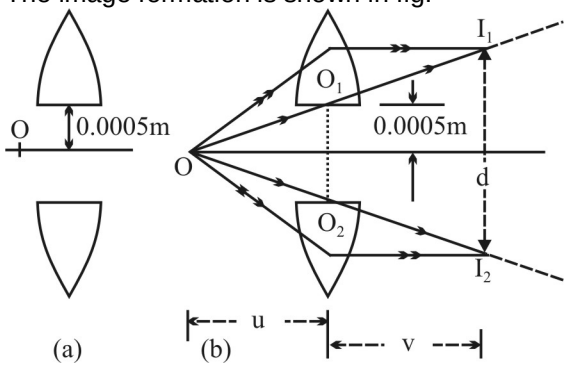
$= 0.6 \times 10^{-6}$ m = 6000 Å

4. D

Sol. Given quantity is = $(a^2 - b^2)\,h$
dimension of h = [L]
dimensions of $a^2 - b^2 = [L^2 - L^2] = L^2$
Therefore the dimensions of the given quantity are $[L^3]$. Thus the quantity being measured is volume
Ans. (D)

5. A

Sol. The image formation is shown in fig.



The rays through optical centres O_1 and O_2 pass undeflected. The image of O are formed at I_1 and I_2 due to upper and lower lenses respectively.

The number of images formed = two

For a convex lens, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

Here u = – 0.3 m and f = 0.2 m

$\therefore \quad \frac{1}{v} = \frac{1}{u} + \frac{1}{f} = -\frac{1}{0.3} + \frac{1}{0.2} = \frac{1}{0.6}$

u = 0.6 m

Let d be the distance between the two images I_1 and I_2. From ΔOO_1O_2 and $\Delta\, II_1I_2$, we have

$$\frac{d}{O_1O_2} = \frac{u+v}{u}$$

or $$\frac{d}{0.0005+0.0005} = \frac{0.3+0.6}{0.3}$$

Solving we get d = 0.003 m

6. C

Sol. Let x be the distance travelled by the truck when both truck and car are side by side. The distance travelled by the car will be
(x + 150) as the car is 150 metre behind the truck. Applying the formula
$s = ut + (1/2)\, a\, t^2$, we have
$x = 1/2 \times (1.5)\, t^2$ .....(1)
and $(x + 150) = (1/2) \times (2)\, t^2$ .....(2)
Here t is the common time.
From eqs. (1) and (2) we have

$$= \frac{x+150}{x} = \frac{2}{1.5}$$

Solving we get x = 450 metre (truck) and
x + 150 = 600 metre (car).
Substituting the value of x in eq. (1),
we get $450 = 1/2\ (1.5)\, t^2$

$\therefore\ t = \sqrt{\frac{450 \times 2}{1.5}} = \sqrt{600} = 24.5$ sec.

Hence correct answer is (C)

7. C

Sol. Velocity of object = 100 km/h.
Velocity of mirror = 40 km/h
Due to velocity of mirror the image will move towards right with double the velocity i.e., 80 km/h. and due to the motion of the object the image will move with velocity 100 km/h towards left. Hence net velocity of the image is 100 – 80 = 20 kmph towards left.
Velocity of image observed to the motorist = 20 + 100 = 120 kmph towards left.
Velocity of image observed to the standing person = 20 kmph.

8. A

Sol. Let us locate the block at an arbitrary angular position $\theta < \theta_o$

$dW = \vec{F} \cdot \vec{dr} = Fds = F(Rd\theta)$ ...(1)

Now from the condition of equilibrium of the block , we have

$F = mg \cos\theta + KR\theta$ ...(2)

from eq. (1) & (2)

$dW = mgR\cos\theta\, d\theta + KR^2\theta\, d\theta$

Hence the sought work

$$W = \int dW = \int_0^{\theta_0} \cos\theta\, d\theta + KR^2 \int_0^{\theta_0} \theta\, d\theta$$

$$W = mgR\sin\theta_0 + \frac{KR^2\theta_0^2}{2}$$

9. A

Sol. According to given data, ionization energy of Li^{++} ion is 122.4 eV

$\therefore$ Excitation energy of this ion

$$= 122.4\left[\frac{1}{1^2} - \frac{1}{2^2}\right]$$

$$= 122.4\left[1 - \frac{1}{4}\right]$$

$$= 122.4 \times \frac{3}{4} \text{ eV}$$

$\therefore$ $E_{ex} = 91.8$ eV

Hence, the value of the first excitation potential

= 91.8 V

10. B

Sol. Moment of inertia of the system about the given axis

$I = I_A + I_B + I_C$

Now as rod is thin

$I_A = \Sigma m \times 0^2 = 0$

Rod B is rotating about one end

$I_B = ML^2/3$

And for rod C all points are always at distance L from the axis of rotation, so

$I_C = \Sigma mL^2 = ML^2$

$\therefore$ $I = 0 + (ML^2/3) + ML^2 = 4ML^2/3$

So if ω is the desired angular speed, gain in kinetic energy due to rotation of H from horizontal to vertical position.

$$K_R = \frac{1}{2} I\omega^2 = \frac{1}{2}\left[\frac{4}{3}ML^2\right]\omega^2$$

And loss in potential energy of the system in doing so

$$= 0 + Mg\frac{L}{2} + MgL = \frac{3}{2} MgL$$

So by conservation of mechanical energy

$(2/3)ML^2\omega^2 = (3/2)MgL$

or $\omega = \frac{3}{2}\sqrt{\frac{g}{L}}$

11. B

Sol. In one half-life the number of active nuclei reduces to half the original number. Thus, in two half-lives the number is reduced to $\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)$ of the original number. The number of remaining active nuclei is, therefore, $6.0 \times 10^{18} \times \left(\frac{1}{2}\right) \times \left(\frac{1}{2}\right) = 1.5 \times 10^{18}$

12. B

Sol. $T = 2\pi \sqrt{\frac{I}{C}}$ .....(1)

$I = \frac{2}{5} MR^2$ .....(2)

By equation (1) & (2)

$T = 2\pi \sqrt{\frac{2MR^2}{5C}}$

given, M = 1kg

R = 0.15 m

$C = 6 \times 10^{-3}$ N-m/rad.

T = ?

$\Rightarrow \quad T = 6.28 \sqrt{\frac{0.4 \times 1 \times (0.15)^2}{6 \times 10^{-3}}}$

$T = 6.28 \sqrt{\frac{9 \times 10^{-3}}{6 \times 10^{-3}}} = 6.28 \times \sqrt{1.5}$

T = 7.7 sec.

13. D

Sol. Moseley's equation is

$\sqrt{\nu} = a (Z - b)$.

Thus $\sqrt{\frac{c}{\lambda_1}} = a(Z_1 - b)$ …(i)

and $\sqrt{\frac{c}{\lambda_2}} = a (Z_2 - b)$ …(ii)

Using equations (i) and (ii), we get

$\sqrt{c}\left(\frac{1}{\sqrt{\lambda_1}} - \frac{1}{\sqrt{\lambda_2}}\right) = a (Z_1 - Z_2)$

$\Rightarrow \quad a = \frac{\sqrt{c}}{(Z_1 - Z_2)}\left(\frac{1}{\sqrt{\lambda_1}} - \frac{1}{\sqrt{\lambda_2}}\right)$

Putting given values

$a = 5.06 \times 10^7$ Hz

Dividing equation (i) by (ii), we get

$$\sqrt{\frac{\lambda_2}{\lambda_1}} = \frac{Z_1 - b}{Z_2 - b}$$

$$\Rightarrow \quad \sqrt{\frac{178.5}{71}} = \frac{42 - b}{27 - b}$$

$$\Rightarrow \quad b = 1.37$$

14. A

Sol. Let the frequency of the fork be n. In the first case, the length of the wire is 70 cm (0.70 m). Therefore,

$$n = \frac{1}{2 \times 0.70}\sqrt{\frac{T}{m}} \qquad ...(1)$$

On decreasing the length of the wire, its frequency will increase. Hence in the second case when the length is 69 cm = 0.69 m, then

$$n + 4 = \frac{1}{2 \times 0.69}\sqrt{\frac{T}{m}} \qquad ...(2)$$

dividing equation (1) by (2), we get

$$\Rightarrow \frac{n}{n+4} = \frac{0.69}{0.70}$$

$$\Rightarrow n \times 0.70 = (n + 4) \times 0.69$$

$$\Rightarrow n = \frac{4 \times 0.69}{0.01} = 276 \text{ sec}^{-1}$$

15. A

Sol. $\tan 60^\circ = \frac{\omega L}{R}$, $\tan 60^\circ = \frac{1/\omega C}{R}$.

$\therefore \omega L = \frac{1}{\omega C}$.

$\therefore$ impedance of the circuit,

$$Z = \left[R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2\right]^{\frac{1}{2}} = R.$$

Current in the circuit,

$$i_0 = \frac{V_0}{Z} = \frac{V_0}{R} = \frac{200}{100} = 2 \text{ ampere.}$$

Average power, $\overline{P} = \frac{1}{2} V_0 i_0 \cos\phi$.

But $\tan\phi = \frac{\omega L - 1/\omega C}{R} = 0$, $\quad \therefore \cos\phi = 1$.

$$\therefore \quad \overline{P} = \frac{1}{2} \times 200 \times 2 \times 1 = 200$$

16. C

Sol. From the law of length of stretched string, we have $n_1 \ell_1 = n_2 \ell_2 = n_3 \ell_3$

Here $n_1 : n_2 : n_3 = 1 : 3 : 15$

$\therefore \frac{\ell_1}{\ell_2}=\frac{n_1}{n_2} = \frac{3}{1}$ and $\frac{\ell_1}{\ell_3}=\frac{n_3}{n_1}$ = 15/1

$\ell_2=\frac{\ell_1}{3}$ and $\ell_3=\frac{\ell_1}{15}$

The total length of the wire is 105 cm.

Therefore $\ell_1 + \ell_2 + \ell_3 = 105$

or $\ell_1 + \frac{\ell_1}{3} + \frac{\ell_1}{15} = 105$ or $\frac{21\ell_1}{15} = 105$

$\ell_1 = \frac{105\times15}{21} = 75$ cm.

$\therefore \ell_2 = \frac{\ell_1}{3} = \frac{75}{3} = 25$ cm

$\ell_3 = \frac{\ell_1}{15} = \frac{75}{15} = 5$ cm

Hence the bridge should be placed at 75 cm and (75 + 25) = 100 cm from one end.

17. B

Sol. We know that

$i = \frac{\varepsilon}{R}(1 - e^{t/\tau})$

$i = \eta\frac{\varepsilon}{R}$ (given)

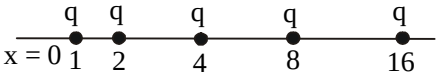
$n\frac{\varepsilon}{R} = \frac{\varepsilon}{R}(1 - e^{-t/\tau})$

or $e^{-t/\tau} = 1 - \eta$

or $t = \tau \ln\left|\frac{1}{1-\eta}\right|$

18. C

Sol. As potential V and intensity E due to a point charge at position x are respectively given by-



$E = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{1^2}+\frac{q}{2^2}+\frac{q}{4^2}+...\right]$

$= \frac{q}{4\pi\varepsilon_0}\left[1+\frac{1}{4}+\frac{1}{16}+\frac{1}{64}....\right]$

i.e.,

$E = \frac{q}{4\pi\varepsilon_0}\frac{1}{[1-(1/4)]} = \frac{1}{4\pi\varepsilon_0}\left[\frac{4}{3}q\right]$

19. C

Sol. The electric field near the plane charge sheet is $E = \sigma/2\varepsilon_0$ in the direction away from the sheet. At the given area, the field is along the Z-axis.

The area = $\pi r^2 = 3.14 \times 1\ cm^2$
$= 3.14 \times 10^{-4}\ m^2$.
The angle between the normal to the area and the field is 60º.

Hence, the flux = $\vec{E}.\Delta\vec{S} = E\Delta S\cos\theta$

$= \frac{\sigma}{2\varepsilon_0} pr^2 \cos 60^\circ$

$= \frac{2.0\times10^{-6}\,C/m^2}{2\times8.85\times10^{-12}\,C^2/N-m^2} \times$

$(3.14 \times 10^{-4} m^2)\ \frac{1}{2}$

$= 17.5\ N\text{-}m^2/C$

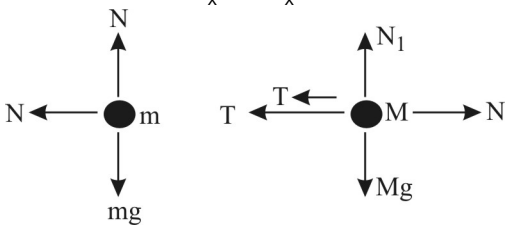
20. A

Sol. Capacitances are -
(A) 10/11 µF (B) 8/13 µF
(C) 6/13 µF (D) 4/11 µF

SECTION – B

21. 5

Sol. Let X' be the leftward displacement of A and x and y be the leftward and downward displacements of m. Then by constraint
$x = X \Rightarrow x = X \Rightarrow a_x = A_x$
and $l_1 - x + l_2 + l_3 - x + l_4 + y$
$= l_1 + l_2 + l_3 + l_4$
where l_1, l_2, l_3, l_4 are the instantaneous lengths of the segments of the string
$\Rightarrow 2x = y \Rightarrow 2x = y \Rightarrow 2a_x = a_y$
$N = ma_x$ and $mg - T = ma_y$
and $2T - N = MA_x = Ma_x$



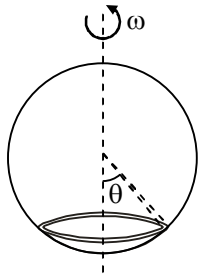
Eliminating T, A_x and N

$a_x = \frac{2mg}{M+5m}$ and $a_y = \frac{4mg}{M+5m}$

$\therefore \quad a = \sqrt{a_x^2 + a_y^2} = \frac{2\sqrt{5}mg}{M+5m}$

22. 3

Sol. When the shell rotates, current is induced due to motion of charge. To calculate magnetic induction at centre of the shell, rotating shell can be assumed to be composed of thin circular current carrying rings. Such a ring can be assumed as follows:
Consider a radius of the shell inclined at angle 'θ' with the axis of rotation. This radius is rotated about the axis keeping θ constant. Thus a circle is traced as shown in fig.



Its radius, $r = R \sin\theta$

Distance of its centre from centre of the shell, $x = R\cos\theta$.

Now consider another radius inclined at angle $(\theta + d\theta)$. It is also rotated in the same way and another circle is traced. The portion between two circles forms a circular ring.

Area of this ring $= 2\pi r\, R\, d\theta = 2\pi R^2 \sin\theta\, d\theta$

Charge on this ring, $dq = \sigma . 2\pi R^2 \sin\theta\, d\theta$

Since, angular velocity of the shell is ω, therefore, it completes $\frac{\omega}{2\pi}$ revolutions per second.

Hence, current associated with the ring considered,

$$i = \frac{\omega}{2\pi} dq = \sigma\omega R^2 \sin\theta\, d\theta$$

Since, centre of the shell is a point lying on the axis of a circular coil of radius r, carrying current i at a distance x from centre of the coil, therefore, magnetic induction at centre of the shell due to this coil is

$$dB = \frac{\mu_0 i r^2}{2(r^2 + x^2)^{3/2}}$$

$$= \frac{1}{2}\mu_0\, \sigma\omega R \sin^3\theta\, d\theta$$

Hence, resultant magnetic induction at centre of the shell

$$B = \int dB = \frac{1}{2}\mu_0\, \sigma\omega R \int_0^{\pi} \sin^3\theta\, d\theta = \frac{2}{3}\mu_0\, \sigma\omega R$$

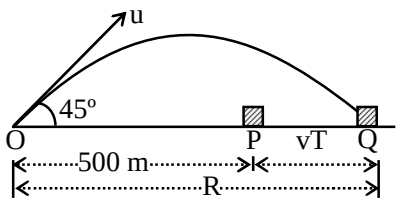
SECTION – C

23. 00088.55

Sol. Consider that at the instant, the shell is fired from the gun at point O, the car is at point P The shell will move along parabolic path for T its time of flight and will then strike the ground at point Q at a distance R, equal to the horizontal range. Therefore, the shell will hit the car, if in the time T (time of flight), the car moves the distance PQ= v T, where v is velocity of the car. Since the car is initially at a distance of 500 m from the gun, it follows that

$$R = 500 + v\,T \quad .....(1)$$

(a) Let u be the speed of projection of shell.



Now, $R = \frac{u^2 \sin 2\theta}{g}$

Here $\theta = 45^o$

$R = \frac{u^2 \sin 2\times 45^o}{9.8} = \frac{u^2 \sin 90^o}{9.8} = \frac{u^2}{9.8}$

Also, $T = \frac{2u\sin\theta}{g} = \frac{2u\sin 45^o}{9.8}$

$= \frac{2u}{9.8} \times \frac{1}{\sqrt{2}} = \frac{\sqrt{2}u}{9.8}$

Further, velocity of the car, v = 72 km h^{-1}
= 72 × (1000m) × (60 × 60 s)$^{-1}$ = 20 m s^{-1}

Substituting for R, T and v in equation (1), we have $\frac{u^2}{9.8} = 500 + 20 \times \frac{\sqrt{2}u}{9.8}$

or $u^2 - \sqrt{2}\,u - 500 \times 9.8 = 0$
On solving for u we have
$u = 10\,(\sqrt{2} + \sqrt{51})$
u = 10 (1.414 + 7.141)
= 10× 8.555 = 85.55 m s^{-1}

24. 00120.00

Sol. As here initially

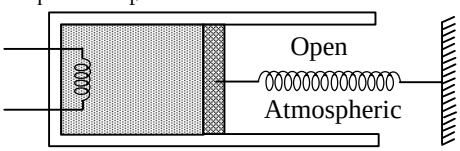
$P_I = P_0 = 1 \times 10^5$ N/m^2,
$V_I = V_0 = 2.4 \times 10^{-3} m^3$,
$T_I = T_0 = 300K$

and finally

$P_F = P_0 + \frac{Kx}{A} = 1 \times 10^5 + \frac{8000 \times 0.1}{8 \times 10^{-3}}$
$= 2 \times 10^5$ N/m^2,

$V_F = V_0 + Ax = 2.4 \times 10 + 0.1 \times 8 \times 10^{-3}$
$= 3.2 \times 10^{-3} m^3$,
So from gas equation PV = μRT, i.e.

$\frac{P_I V_I}{T_I} = \frac{P_F V_F}{T_F}$



we have

$T_F = \frac{P_F}{P_I} \times \frac{V_F}{V_I} \times T_I$

$= \frac{2\times 10^5}{1\times 10^5} \times \frac{3.2\times 10^{-3}}{2.4\times 10^{-3}} \times 300 = 800$ K

As $\Delta W = \int P dV$,
here $P = P_0 + (kx/A)$

so $\Delta W = \int_0^{0.1} \left(P_0 + \frac{kx}{A}\right) A\, dx$

$$= \int_0^{0.1} (P_0A + kx)dx$$

i.e. $= P_0Ax + ½ kx^2$

$= [10^5 \times 8 \times 10^{-3} \times 0.1 + ½ \times 8000 \times (0.1)^2]$

i.e. $\Delta W = 80 + 40 = 120J$

25. 00680.00

Sol. The observer A is at rest with respect to the air and the source is travelling at a velocity of 120 km/h i.e., 100/3 m/s. As is clear from the figure, the person receives the sound of the whistle in a direction BA making an angle θ with the track where $\cos\theta = 300/500 = 3/5$. The component of the velocity of the source (i.e., of the train.) along with this direction AB is $\frac{100}{3} \times \frac{3}{5}$ m/s = 20m/s. As the source is approaching the person with this component, the frequency heard by the observer is.

$$\nu' = \frac{v}{v - v\cos\theta}\nu = \frac{340}{340 - 20} \times 640\text{Hz} = 680 \text{ Hz}.$$

Chemistry

PART – II

SECTION – A

26. A

27. C

28. B

Sol. $Na\left[\overset{+I}{A}g\left(CN\right)_2\right]$

29. A

30. D

Sol. $EDTA^{4-}$ is Hexadentate and form optically active compounds.

31. C

32. B

33. C

Sol. Rate of effusion $\propto \frac{1}{\sqrt{T}}$

34. A

35. B

Sol. Area under the curve $= -1\times3\times10^{-5} - \frac{1}{2}\times3\times3\times10^{-5} = -7.5\times10^{-5}\ J$

36. C

Sol. $P_4 + 20HNO_3 \longrightarrow 4H_3PO_4 + 20\,NO_2\uparrow + 4H_2O$

$m.moles\ of\ NaOH = 2\times m.moles\ of\ H_3PO_4 \Rightarrow m.moles\ H_3PO_4 = \frac{100\times0.1}{2} = 5$

$m.moles\ of\ HNO_3 = \frac{20}{4}\times5 = 25$

Volume of $HNO_3 = 25\,ml$

37. B

38. B

39. A

40. C

41. C
Sol. Zn^{2+}, Li^{+}, K^{+}, Al^{3+} are colourless where as Co^{2+} is coloured.

42. A

43. B

44. C

45. D

SECTION – B

46 4
Sol. d,e,f,g are correct and Colloid is +ve.

47. 4

Sol. $\text{Volume of unit cell}=\frac{\sqrt{3}}{4}\times(4.53)^2\times7.41\times10^{-24}\text{cm}^3$

$\text{so mass}=60.57\times10^{-24}\text{g}$

$\text{moles}= 3.365\times10^{-24}$

$\text{molecules of } H_2O=2.025$

$\text{number of H atom}=4$

SECTION – C

48. 00027.00

49. 00039.00
Sol. X=120 (sp^2-hybridization)
Y=180 (sp-hybridization)
Z=90 (sp^3d^2-hybridization)

$$\frac{120+180+90}{10}=\frac{390}{10}=39$$

50. 00013.00

Mathematics

PART – III

SECTION – A

51. B

Sol. $\int \frac{3x^2+4x-1}{\left(x^2+1\right)^2\sqrt{x+1}}dx$

$= \int \frac{4x^2+4x-x^2-1}{\left(x^2+1\right)^2\sqrt{x+1}}dx$

$= 4\int \frac{x\left(x+1\right)dx}{\left(x^2+1\right)^2\sqrt{x+1}} - \int \frac{dx}{\left(x^2+1\right)\sqrt{x+1}}$

$4\int \frac{x\left(x+1\right)dx}{\left(x^2+1\right)^2\sqrt{x+1}}.\left[\frac{2\sqrt{x+1}}{x^2+1}\int \frac{-1 \quad 2x.2\sqrt{x+1}}{\left(x^2+1\right)^2}.dx\right]$

$= 4\int \frac{x\left(x+1\right)dx}{\left(x^2+1\right)^2\sqrt{x+1}} - \frac{2\sqrt{x+1}}{x^2+1}$

$= \frac{-2\sqrt{x+1}}{x^2+1} + c$

52. C

Sol. $y\left(xy-\left(x^2-y^2\right)^2\right).dx = \left\{y^3 - x\left(x^2-y^2\right)^2\right\}dy$

$\Rightarrow xy^2dx - y\left(x^2-y^2\right)^2 dx = y^3dy - x\left(x^2-y^2\right)^2 dy$

$\Rightarrow y^2\left(xdx-ydy\right) = \left(x^2-y^2\right)^2\left(ydx-xdy\right)$

$\Rightarrow xdx-ydy = \left(x^2-y^2\right)^2\frac{\left(ydx-xdy\right)}{y^2}$

$\frac{2\left(xdx-ydy\right)}{\left(x^2-y^2\right)^2} = \frac{2\left(ydx-xdy\right)}{y^2}$

$\int \frac{d\left(x^2-y^2\right)}{\left(x^2-y^2\right)^2} = 2\int d\left(\frac{x}{y}\right)$

$\frac{-1}{x^2-y^2} = \frac{2x}{y} + C$

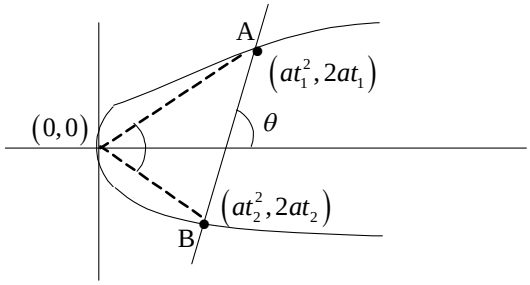
$\therefore \frac{2x}{y} + \frac{1}{x^2-y^2} = C$

53. D

Sol. $y^2 = 8x$

Now, $t_1 t_2 = -4$

And $t_2 = -t_1 - \frac{2}{t_1}$



$\frac{-4}{t_1} = -t_1 - \frac{2}{t_1}$

$\Rightarrow -t_1^2 = -2$

$\Rightarrow t_1^2 = 2$

$y^2 = 4ax$

$\frac{dy}{dx} = \frac{4a}{2y}$

$\frac{dy}{dx} = \frac{1}{t}$

Slope of normal $= -t$

And $\tan^2\theta = t_1^2 = 2$

54. A

Sol. $C : \left(x^2y^2 - 4x - 8y - 5\right) + \lambda\left(x^2 + y^2 - 12x + 2y + 29\right) = 0$

Centre : $\left(\frac{2+6\lambda}{1+\lambda}, \frac{4-\lambda}{1+\lambda}\right)$

Common chord $4x - 5y - 17 = 0$

Centre lies on common chord

$\therefore \lambda = \frac{29}{12}$

$\therefore$ centre $\left(\frac{1+6(29/12)}{1+29/12}, \frac{4-29/12}{1+29/12}\right)$

$\therefore$ centre $\left(\frac{198}{41}, \frac{19}{41}\right)$

55. B

Sol. $\int_0^{\frac{11\pi}{3}} \sqrt{1+\cos 2x}.dx$

$= \int_0^{\frac{11\pi}{3}} \sqrt{2}\left|\cos x\right|.dx$

$= \int_0^{3\pi} \left|\sqrt{2}\cos x\right|.dx + \int_{3\pi}^{3\pi+2\pi/3} \sqrt{2}\left|\cos x\right| dx$

$3\sqrt{2}\int_0^{\pi} \left|\cos x\right|.dx + \sqrt{2}\int_0^{2\pi/3} \left|\cos x\right|.dx$

$= 6\sqrt{2}\int_0^{\pi/2} \left(\cos x\right) dx + \sqrt{2}\int_0^{\pi/2} \cos x.dx - \sqrt{2}\int_{\pi/2}^{2\pi/3} \cos x.dx$

$= 6\sqrt{2} + \sqrt{2}\,\frac{-\sqrt{3}}{\sqrt{2}} + \sqrt{2}$

$= \sqrt{2} - \sqrt{\frac{3}{2}}$

56. B

Sol. $2\vec{a} - \vec{b} = \hat{i} + 3\hat{j} - 5\hat{k}$

$\vec{a} - 2\vec{b} = -4\hat{i} + 4\hat{j} - 4\hat{k}$

Area of $= \frac{1}{2}\left|\left(2\vec{a} - \vec{b}\right) \times \left(\vec{a} - 2\vec{b}\right)\right|$

$\left(2\vec{a} - \vec{b}\right) \times \left(\vec{a} - 2\vec{b}\right) = 3\hat{i} + 24\hat{j} + 15\hat{k}$

Area $= \frac{9\sqrt{10}}{2}$

57. D

Sol. Taking first two

$\frac{\tan\left(\pi/4 + \alpha\right)}{\tan\left(\pi/4 + \beta\right)} = \frac{5}{3}$

$\therefore \frac{\cos\left(\alpha + \beta\right)}{\sin\left(\alpha - \beta\right)} = \frac{8}{2} = 4$

$\therefore \cos\left(\alpha + \beta\right) = 4\sin\left(\alpha - \beta\right) - - - - - - - (1)$

Similarly

$\cos\left(\beta + \alpha\right) = \sin\left(\beta - \alpha\right) - - - - - - (2)$

$3\cos\left(\gamma + \alpha\right) = -7\sin\left(\gamma - \alpha\right) - - - - - (3)$

After marking $\sin^{-2}$ terms

$(1) + (2) - (3)$

Then,

$12\sin^2\left(\alpha - \beta\right) + 15\sin^2\left(\beta - \gamma\right) - 7\sin\left(\gamma - \alpha\right) = 0$

58. A

Sol. At $x=0, y=1$ for both curves intersect

Now $y=(1+x)^{\cos x}+\sin x$

$\therefore \frac{dy}{dx}=2 \; at \; x=0$

And $y=\frac{1}{2}\left(x^2+x+2\right)$

$\frac{dy}{dx}=\frac{1}{2} \; at \; x=0$

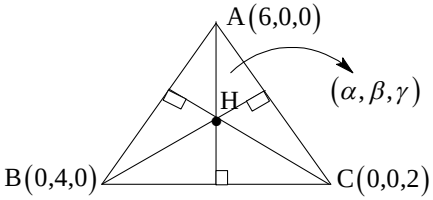
Now $m_1=2, \; m_2=\frac{1}{2}$

$\therefore \tan\theta=\left|\frac{2-1/2}{1+(2)(1/2)}\right|=\left|\frac{3/2}{2}\right| \Rightarrow \frac{3}{4}$

59. A

Sol. Now $AH \perp BC$

Now D^r of $AH=(\alpha-6,\beta,\gamma)$



$D^r \; of \; BC=(0,4,-2)$

Now for 1 lines,

$a_1a_2+b_1b_2+c_1c_2=0$

$\therefore 0(\alpha-6)+\beta(4)-2\gamma=0$

$\therefore 4\beta-2\gamma=0$

$2\beta=\gamma---(1)$

Similarly

$-6\alpha+2\gamma=0-----(2)$

$-6\alpha+4\beta=0-----(3)$

$\therefore 2\beta=\gamma; \; \frac{2\beta}{3}=\alpha$

Now may lie on

$2x+3y+6z=12$

$\therefore 2\alpha+3\beta+6\gamma=12$

$2\left(\frac{2\beta}{3}\right)+3\beta+6(2\beta)=12$

$4\beta+9\beta+36\beta=36$

$49\beta=36$

$\beta=\dfrac{36}{49}$

Now, $\alpha=\dfrac{24}{49}$ and $\gamma=\dfrac{72}{49}$

$(\alpha,\beta,\gamma)=\left(\dfrac{24}{49},\dfrac{36}{49},\dfrac{72}{49}\right)$

60. C

Sol. $f(x)=\log_3\left(\log_{1/2}\left(\log_2\left(\log_{1/3}\right)\right)\right)$

$\dfrac{x+1}{\sqrt{1-3x-4x^2}}$

Now, $\sqrt{1-3x-4x^2}>0$

$4x^2+3x-1<0$

$\therefore (4x-1)(x+1)<0$

$\therefore x\in(-1,1/4)$

$\log_{1/2}\left(\log_2\left(\log_{1/3}(x)\right)\right)>0$

$\log_2\left(\log_{1/3}(x)\right)<1$

$\log_{1/3}(x)<2$

$x>1/9$

Now $F\left(f(x)\right)\in x>\dfrac{1}{9}\cap x\in(-1,1/4)$

$\therefore x\in\left(\dfrac{1}{9},\dfrac{1}{4}\right)$

61. D

Sol. $f(x)+f(1-x)=\dfrac{1}{27}$

$\dfrac{54}{27}=2$

62. C

Sol. $\cot\sum_{n=1}^{\infty}\cot^{-1}\dfrac{1}{n^2}\times n^2\,\dfrac{(n+1)^2}{2}$

$$= \cot \sum_{n=1}^{\infty} \tan^{-1} \frac{2}{(n+1)^2}$$

$$= \cot \sum_{h=1}^{\infty} \left(\tan^{-1}(n+2) - \tan^{-1} n \right)$$

$$s_n = \begin{bmatrix} \tan^{-1} 3 - \tan^{-1} 1 \\ \tan^{-1} 4 - \tan^{-1} 2 \\ \tan^{-1}(n+2) - \tan^{-1} n \end{bmatrix}$$

$$s_n = \tan^{-1}\left(\frac{3n^2 + 7n}{n^2 + 9n + 10} \right)$$

$$s_\infty = \tan^{-13}$$

$$\cot \tan^{-13} = \cot \cot^{-1} \frac{1}{3} \Rightarrow \frac{1}{3}$$

63. A

Sol. $$\lim_{x \to \pi} a \frac{\left\{ \frac{1}{1+9f(x)} \right\} 9f'(x) - \cos(f(x)) f'(x)}{2f'(x)}$$

$$= \frac{9-1}{2} = 4$$

$$\sum_{h=1}^{\infty} 4^{-n} = \frac{1}{4} + \frac{1}{4^2} + ...\infty$$

$$= \frac{1}{3}$$

64. C

Sol. Centre $(0, -f)$

Equation of circle

$$x^2 + y^2 + 2fy + c = 0$$

$$\left(\log_e a_r \right)^2 + \frac{1}{\left(\log_e a_r \right)^2} + 2f\left(\frac{1}{\log_e a_r} \right) + c = 0$$

$$\left(\log a_r \right)^4 + c\left(\log a_r \right)^2 + 2f\left(\log a_r \right) = 0$$

$\log a_1, \log a_2, \log a_3, \log a_4$ are roots of equation

$$\therefore \log\left(a_1 . a_2 . a_3 . a_4 \right) = 0$$

$$a_1 a_2 a_3 a_4 = 1$$

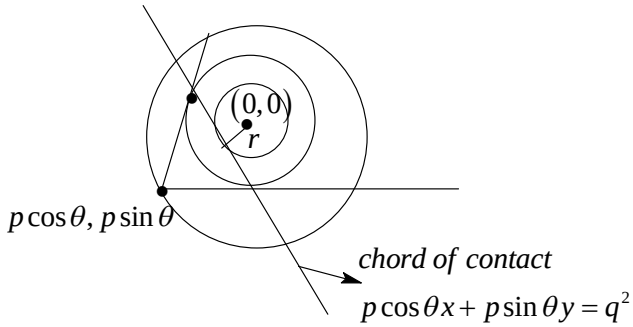
$$\frac{a_1 + a_2 + a_3 + a_4}{4} \geq (1)^{1/4}$$

$\therefore a_1 + a_2 + a_3 + a_4 \geq 4$

65. A

Sol. $\perp$ distance from (centre) to chord of contact

$$r = \left| \frac{p\cos\theta \times 0 + p\sin\theta \times 0 - q^2}{p} \right|$$



$$q^2 = rp$$

$p, q, r \text{ are in } GP$

$\therefore 2a - \log_e^p,\ 4a - \log_e^q,\ 6a - \log_e^r \text{ are in } AP$

66. D

Sol. *G.E.of line*

$Ax + By + C = 0$

$At_1 + B\left(2at_1 + at_1^3\right) + c = 0$

$Bat_1^3 + \left(2aB + A\right)t_1 + c = 0$

Sum of roots

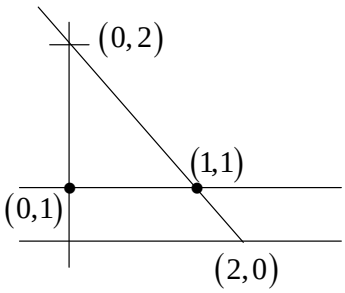
$t_1 + t_2 + t_3 = 0$

$2^{t_1}.2^{t_2}.2^{t_3} = 2^\circ = 1$

67. D

Sol. $x^2 - y^2 + 2y - 1 = 0$

$x^2 - \left(y-1\right)^2 = 0$



$\left(x + y - 1\right)\left(x - y + 1\right) = 0$

Angle bisector

$$\frac{x+y-1}{\sqrt{2}}=\pm\frac{x-y+1}{\sqrt{2}}$$

$$x+y-1=x-y+1$$

$$x+y-1=-x+y-1$$

$$\Rightarrow y=1 \quad \Rightarrow x=0$$

Area $a=\frac{1}{2}\times1\times1=\frac{1}{2}$

Orthocentre $(0,1)\Rightarrow b=0,c=1$

$$\therefore 4a+b+c=2+0+1=3$$

68. A

Sol. $z^5=1$

$$\frac{z^5}{z^5+z}+\frac{z}{z+1}+\frac{z^2}{z^2+1}+\frac{z^5}{z^2+z^5}$$

$$=\frac{1}{1+z}+\frac{z}{1+z}+\frac{z^2}{z^2+1}+\frac{1}{z^2+1}$$

$$=1+1=2$$

69. D

Sol. Equation of tangent $y=3x\pm\sqrt{9a^2+b^2}$

This is normal to circle

$$x^2+y^2+4x+1=0$$

Panes – 1 brought $(-2,0)$

$$\Rightarrow 0=-6\pm\sqrt{9a^2+b^2}$$

$$36=9a^2+b^2$$

$$AM\geq GM$$

$$\frac{9a^2+b^2}{2}\geq\sqrt{9a^2b^2}$$

$$18\geq 3ab$$

$$6\geq ab$$

70. C

Sol. $F(x)=\int_0^x f(t)dt$

$$F(x^2)=\int_0^{x^2} f(t)dt$$

$$=x^2(1+x)$$

Diff. both side

$$F'(x^2).2x=2x+3x^2$$

$f\left(x^2\right)=1+\frac{3}{2}x$

$x=2,\ f\left(4\right)=1+\frac{3x}{2}$

$x=2,\ f\left(4\right)=1+\frac{3}{2}\times 2=4$

Unit place $\left(985697\right)^{16}=1$

SECTION – B

71. 2

Sol. $1920=5\times 3\times 2^7=5\times 3\times 2\times 2\times 2\times 2\times 2\times 2\times 2=\frac{9!}{7!}=9\times 8$

$1920=5\times 3\times 4\times 2\times 2\times 2\times 2\times 2\times 1=\frac{9!}{5!}=9\times 8\times 7\times 6$

$1920=3,5,4,4,2,2,2,1,1$

$\frac{9!}{2!2!3!}$

$3,5,4,4,4,2,1,1,1$

$\frac{9!}{3!3!}$

Total = $\frac{9!}{7!}+\frac{9!}{5!}+\frac{9!}{2!2!3!}+\frac{9!}{3!3!}$

$=\frac{9!}{7!}\left(1+42+210+140\right)$

$=393\ {}^{9}\mathrm{P}_{2}$

72. 5

Sol. $20c_0\times 30c_{5m}+20c_1\times 30c_{5m-1}$

$+.....+20c_{5m}\times 30c_0$

Coeff of $x^{5m}\rightarrow\left(1+x\right)^{20}\left(1+x\right)^{30}$

$50c_{5m},5m=\frac{50}{2}=25$

$m=5$

SECTION – C

73. 00012.00

Sol. $f\left(x\right)+f\left(x+2\right)=\sqrt{3}f\left(x+1\right)$

$x\rightarrow x+2------\left(1\right)$

$f(x+2)+f(x-14)=\sqrt{3}f(x+3)------(2)$

$(1)+(2)$

$f(x)+2+(x+2)+f(x+14)$

$=\sqrt{3}\{f(x+1)+f(x+3)\}$

$=3f(x+2)$

$f(x)+f(x+4)=f(x+2)$

$x\to x+2-----(3)$

$f(x+2)+f(x-16)=f(x+4)---(4)$

$(3)+(4)$

$f(x)+f(x+6)=0$

$f(x+6)=-f(x)$

$f(x+12)=-f(x+6)$

$=f(x)$

Period $T=12$

74. 00019.00

75. 00081.00

Sol. Given $\frac{f(x)f'(x)}{\sqrt{1-(f(x))^4}}\geq x$

$\frac{2f(x)f'(x)}{\sqrt{1-(f(x))^4}}-2x\geq 0$

Let $g(x)=\sin^{-1}(f(x))^2-x^2$ is non decreasing function

$\lim_{x\to a^-}g(x)\leq\lim_{x\to b^+}g(x)$

$\frac{\pi}{2}-a^2\leq\frac{\pi}{6}-b^2$

$a^2-b^2\geq\frac{\pi}{3};\left[a^2-b^2\right]=1$